Date: Oct 01, 2025

Math Trimester 1

MorningWork

Name:
Class of:
Teacher:
Homeroom:

Multiple Choice — Quadratic Functions

1) Which of the following is a quadratic function?

A. $f(x) = 3x + 5$

B. $f(x) = x^2 - 4x + 1$

C. $f(x) = \sqrt{x} + 2$

D. $f(x) = \frac{4}{x}$

2) The graph of $y = (x-3)^2 - 7$ has its vertex at:

A. $(3, 7)$

B. $(-3, -7)$

C. $(3, -7)$

D. $(-3, 7)$

3) What are the zeros (x-intercepts) of $y = x^2 - 5x + 6$?

A. $x = -2,\ -3$

B. $x = 2,\ 3$

C. $x = -2,\ 3$

D. $x = -1,\ 6$

4) For the quadratic $y = -2(x+1)^2 + 9$, the parabola:

A. opens upward and has a minimum at $y = 9$

B. opens upward and has a maximum at $y = 9$

C. opens downward and has a maximum at $y = 9$

D. opens downward and has a minimum at $y = 9$

5) Consider $y = ax^2 + bx + c$. If the discriminant $b^2 - 4ac$ is negative, which statement is true?

A. The graph has two distinct real x-intercepts.

B. The graph has one real x-intercept (a repeated root).

C. The graph has no real x-intercepts.

D. The graph is not a parabola.